Targets with an Error Model

Joint distributions, paired errors, and conditional requirements

M: observations. Blue **B**: conditional error requirements. **T**: goals. B is evaluator-side accounting, not a deployable strategy or future forecast.

E1 / body Table 3: original-order comparison and spread

Each supported cell contains 4,710 records from **157 unique molecules**, five fitted priors, two seed labels, and three delays. SD is the spread of five fit-fold means; the same cohort and arrival order recur.

Table 1: E1. Accuracy (%), descriptive fit SD (pp), and error counts. SD is not a standard error.

	Natural			Controlled		
Method	Acc.	SD	Errors	Acc.	SD	Errors
Random M	83.76	0.00	765	—	—	—
Fit-only Best Fixed M	88.15	0.57	558	—	—	—
Frozen M	89.81	1.19	480	73.38	5.83	1254
Online SGD M	89.21	1.01	508	84.03	5.98	752
LinUCB M	89.72	1.11	484	81.15	5.25	888
BayesianRouter M	89.53	0.84	493	80.28	2.64	929
Marginal reliability M	89.81	1.19	480	—	—	—
MIRA-Relay M	89.81	0.21	480	87.56	1.86	586
Retrospective Best Fixed M	90.45	0.00	450	—	—	—

MIRA, Frozen, SGD and LinUCB seed 0/1 vectors are identical. Random's zero fit SD reflects reused decisions, not zero uncertainty. The fixed-provider control is not the per-molecule oracle.

Table 2: E1. Paired error changes relative to the same Frozen decisions (M).

Condition	Method	Rescued	Harmed	Both wrong	Net (pp)
Natural	Online SGD	84	112	396	-0.59
Natural	LinUCB	24	28	456	-0.08
Natural	BayesianRouter	92	105	388	-0.28
Natural	MIRA-Relay	82	82	398	+0.00
Controlled	Online SGD	720	218	534	+10.66
Controlled	LinUCB	414	48	840	+7.77
Controlled	BayesianRouter	492	167	762	+6.90
Controlled	MIRA-Relay	780	112	474	+14.18

Rescued = method correct / Frozen wrong; harmed = method wrong / Frozen correct. Natural Relay's 82 rescues and 82 harms cancel. Controlled Relay's 780 rescues exceed its 112 harms. Neutral action changes are not rescues.

Joint provider difficulty: preserve the actual dependence

Table 3: Native joint correctness vectors (M): one bit per provider, 1 = correct.

Bits	Dev.	Fresh	Bits	Dev.	Fresh	Bits	Dev.	Fresh	Bits	Dev.	Fresh
0000	104	7	0100	3	1	1000	20	3	1100	13	3
0001	8	3	0101	2	0	1001	0	0	1101	1	0
0010	1	0	0110	0	0	1010	2	1	1110	65	21
0011	15	1	0111	9	7	1011	8	0	1111	471	110

Bit order: logreg / MLP / random forest / graph. Development: 471 all-correct, 147 mixed, 104 all-wrong. Fresh: 110 / 40 / 7. Native oracle: 618/722 and 150/157. Independent provider-error draws would distort this structure. Controlled flips require a separate panel: its recorded oracle is 4572/4710.

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E1–E2 / body Table 4 and Figure 3: one shared budget

Let c be a recorded Relay correctness bit, h indicate Relay wrong/Frozen correct, and r indicate Relay correct/Frozen wrong. A conditional repair budget is

$$c' = c + \alpha h - \beta r, \qquad 0 \le \alpha, \beta \le 1.$$

Case: for a harmed decision $(c,h,r) = (0,1,0)$, $\alpha = .6$ gives fractional score .6; for a rescued decision $(1,0,1)$, $\beta = .02$ gives .98. These are budget weights, not binary predictions. Assumption: the same fractions apply uniformly within every observed error stratum, in both conditions. All other records are unchanged. Altered future feedback would require a full causal policy run.

Table 4: E1 conditional response surface (B), including null and adverse outcomes.

			Natural		Controlled	
Scenario	α(%)	β(%)	Acc.(%)	Δ(pp)	Acc.(%)	Δ(pp)
Observed M	0.00	0.00	89.81	+0.00	87.56	+0.00
Loss only B	0.00	10.00	89.63	-0.17	85.90	-1.66
Blunt repair B	30.00	10.00	90.16	+0.35	86.62	-0.94
Selective repair B	60.00	2.00	90.82	+1.01	88.65	+1.10
v05 E1 inverse B	59.67	1.13	90.83	+1.02	88.79	+1.23

Delta is relative to recorded Relay, not Frozen. The v05 inverse solves exactly for its former +48/+58 gains: prevent 59.67% of harms while losing only 1.13% of rescues. These demanding selectivity requirements have not been demonstrated. B has no target CI or new seed distribution.

Table 5: E1 stratum profiles from the same ledger: M and selective-repair B.

	Natural accuracy (%)		Controlled accuracy (%)	
Stratum	M	B	M	B
delay 0	89.68	90.72	87.01	88.14
delay 1	89.68	90.79	87.90	89.09
delay 4	90.06	90.95	87.77	88.74
fold 0	90.02	91.90	89.38	90.62
fold 1	89.60	89.83	84.93	86.03
fold 2	89.60	90.83	86.41	87.32
fold 3	90.02	90.36	88.96	89.30
fold 4	89.81	91.16	88.11	90.00

Fold = fitted-prior condition, not a separate confirmation sample. Delay counts are 1570 each; fold counts are 942 each. Natural delay means are nonmonotone and remain so; no curve was smoothed to imply that shorter delay always helps. Conditional fit SD is a spread of transformed recorded strata, not predictive uncertainty.

Table 6: E2 weighting consistency, computed from the E1 molecule ledger.

Condition	Status	Pooled–F	Scaffold–F	Scaffold–old Perm.
Natural	Recorded M	+0.00	+0.10	-0.30
Natural	v05 E1 inverse B	+1.02	+0.84	+0.44
Controlled	Recorded M	+14.18	+14.40	+15.67
Controlled	v05 E1 inverse B	+15.41	+15.30	+16.57

All contrasts are pp. v05 E1 means imply +0.44pp against the old natural permutation, not its separately chosen +1.00pp. A future attribution result MUST rerun that candidate's own matched permutation; holding the old comparator fixed is only budget accounting, not a causal estimate of the new mechanism.

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E3 / body Table 5: cold-start errors, not just a mean

The development OOF folds partition 722 distinct molecules. Across the native panel, only mixed molecules can change correctness when the provider menu is fixed. For this cohort,

$$\text{accuracy} = \frac{471 + 147q}{722},$$

Case: q is mixed-molecule selection accuracy. Corrected Frozen has $q = 108/147 = 73.47\%$, giving 579/722=80.19%. Fresh has a different equation $(110+40q)/157$; Relay's $q = 77.50\%$ gives 89.81%. The cohort mix explains much of the level difference, not superior adaptation alone.

Table 7: E3 measured error composition by held-out development fold (M).

Fold	Molecules	Correct	All wrong	Avoidable	Mixed correct
0	145	108	28	9	21/30
1	145	111	25	9	25/34
2	144	119	19	6	19/25
3	144	125	12	7	19/26
4	144	116	20	8	24/32

All-wrong errors cannot be fixed by selection. Corrected has only 39 avoidable unique errors: 9/9/6/7/8. Its remaining rank gaps do not calibrate a probability of rescue.

Table 8: E3 comparable prior attempts: actual paired negative results (M).

Candidate	Acc.(%)	Rescued	Harmed	Fold net counts
BCE/seed0	78.53	16	28	-6, -2, -3, +1, -2
BCE/seed1	78.95	17	26	+2, -5, +0, +1, -7
BT/seed0	78.81	14	24	-7, +1, -2, -1, -1
BT/seed1	78.25	9	23	+1, -6, -1, -1, -7

All recorded leave-one-fold-out effects are negative. Do not replace these with two tightly spaced winning future seeds. Future training-seed uncertainty is uncalibrated.

E4 / body Table 6: keep prior and update effects separate

Table 9: E4 measured prior by update factorial (M), 2166 records per cell.

	Natural		Controlled	
Prior + update	Acc.(%)	Errors	Acc.(%)	Errors
raw + frozen	79.36	447	69.11	669
raw + marginal	79.18	451	77.29	492
raw + query	78.49	466	75.95	521
corrected + frozen	80.19	429	63.30	795
corrected + marginal	79.82	437	78.30	470
corrected + query	80.33	426	78.67	462

Corrected/query vs corrected/frozen: natural rescue/harm=26/23 (net+3), controlled=425/92 (net+333). A better prior cannot count as update benefit. Controlled provider flips change the joint error partition; the native 471/147/104 partition must not be reused there.

Table 10: Goals translated into error requirements (T), not candidate forecasts.

Component	Goal	Added correct	Required mixed success	Minimum requirement
E3 prior T	591/722	+12	120/147 = 81.63%	12 net rescues from 39
E4 query T	1770/2166	+30	357/441 = 80.95%	30 net rescues from 114

E3/E4 have different comparators; these gains cannot be added into a future E1 result. BCE seed0 needs at least 24/28 harmful switches repaired to reach 591 while retaining its 16 rescues. Query has only 23 harms: fixing all reaches 1763, so at least seven additional rescues are needed for 1770. Of 114 avoidable query errors, 109 had a released correct-provider receipt. This is provider-level coverage, not local support or a bound on earlier-release benefit. New prior, own Frozen, new update and their combination need a common evaluation contract.

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E5 / body Table 7: matched cost and feasibility

Costs are the mean of 30 recorded stream-level p95 kernel times on macOS arm64. They exclude encoding, queue, I/O and end-to-end tool time. They do not describe A800 performance.

Table 11: E5 measured kernel costs (M) and research caps (T), microseconds.

	Natural		Controlled	
Method	Select	Update	Select	Update
Frozen M	15.89	1.31	17.88	1.50
Online SGD M	14.08	19.37	14.29	20.02
LinUCB M	128.63	90.90	128.36	92.32
BayesianRouter M	133.65	88.96	137.34	92.55
MIRA-Relay M	27.21	2.81	32.47	3.46
Candidate cap T	≤ 30	≤ 5	≤ 35	≤ 5

T is a feasibility requirement, not an estimated time. More capacity or a changed representation requires real remeasurement. Live peak memory and end-to-end latency have not been measured; no target SD is imputed.

E6 / body Table 8: coverage must match within each fitted policy

Table 12: E6 actual fit-level coverage and available hindsight swaps (M).

Fit	Accepted	Wrong	Refused	Refused correct	Swap cap	Acc.(%)
0	119	5	38	26	5	95.80
1	134	7	23	11	7	94.78
2	137	8	20	9	8	94.16
3	137	8	20	9	8	94.16
4	136	8	21	10	8	94.12

785 records reuse 157 molecules under five fitted policies. Accepted counts 119/134/137/137/136 must each remain fixed in a matched-coverage comparison. Refused correctness is obtained from the matching always-accept evaluator ledger, not imputed from null answer fields. Swap caps use truth and do not prove a legal gate can rank them.

Table 13: E6 same-coverage conditional error and utility budget (B).

Status	Wrong accepted	Accepted acc.(%)	Utility
Recorded M	36	94.57	-0.0947
4 favorable swaps B	32	95.17	-0.0896
8 favorable swaps B	28	95.78	-0.0845

Each favorable swap replaces an accepted-wrong with a refused-correct output within the same fit; adverse swaps subtract from the gain. Eight net swaps are 22.22% of current accepted errors. Their allocation and ranking quality must be observed, not invented. Utility=-(W+.25R+.01C)/N, with C=N=785. New routing changes these outputs and requires a new gate evaluation.

A joint success criterion

A scientifically useful improvement should reduce harmful switches while retaining drift recovery, beat its *own* Frozen and matched mechanism controls, and satisfy the kernel budget at the same time. E6 remains a separately frozen gate unless its joint routing outputs are evaluated. Six independently improved means do not establish one successful candidate.

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E1–E2: measured distribution over 128 arrival orders

These **M** results were completed by the parallel experimental work and independently verified. One independent uniform permutation of the fixed 157-molecule cohort is one unit, averaging the same 30 fixed fit/seed/delay configurations. Methods share each order and execute full adaptive queues. The original order is excluded. This is *order uncertainty on a fixed cohort*, not uncertainty over new molecules or future candidates. Stress flips the fit-best provider from tick 78 in each order; which molecules receive the intervention changes with that order.

Table 14: Order-mean accuracy and actual order SD (M); all delays averaged.

	Natural		Controlled	
Method	Mean(%)	SD(pp)	Mean(%)	SD(pp)
Random M	86.47	1.15	—	—
Fit-only Best Fixed M	88.15	0.00	—	—
Frozen M	89.81	0.00	75.75	1.46
Online SGD M	89.05	0.61	82.22	2.87
LinUCB M	89.70	0.33	76.90	2.48
BayesianRouter M	89.72	0.57	79.60	1.64
Marginal reliability M	89.80	0.17	—	—
MIRA-Relay M	89.52	0.49	86.96	1.46
Credit permutation M	89.86	0.46	71.40	3.72

These are order SDs, not the fit SDs on page 1. Natural Frozen is constant across orders. Natural MIRA mean is lower than Frozen; controlled recovery remains positive. Do not assume an improved future method will inherit these SDs.

Table 15: Actual paired effect distributions (M), in pp; quantiles are not mean CIs.

Matched contrast	Mean	SD	q025	q975	Wins
Natural: MIRA–Frozen	-0.29	0.49	-1.18	+0.54	36/128
Controlled: MIRA–Frozen	+11.21	2.46	+6.68	+15.68	128/128
Controlled: MIRA–SGD	+4.74	2.40	+0.45	+8.61	125/128
Controlled: Local–Perm.	+15.57	4.39	+6.05	+23.31	128/128

A positive central order quantile does not mean every order wins: MIRA–SGD ranges from -2.68 to +10.28pp, with 3 negative orders. The Local–Perm. row is pooled correctness, not the equal-scaffold estimand on page 2.

Table 16: Cross-method dependence under controlled drift (M).

MIRA comparator	Cov.(pp^2)	Correlation	Paired SD(pp)	If independent(pp)
SGD	+2.31	+0.55	2.40	3.22
Frozen	-0.91	-0.43	2.46	2.06
Permutation	-1.66	-0.31	4.39	4.00

The last column is a counterexample to an incorrect independence assumption, not a valid error estimate. For natural Frozen, correlation is undefined because its vector is constant.

$$s'^2_{A-B} = s'^2_A + s'^2_B - 2\,\mathrm{Cov}(A,B).$$

Case: A and B are MIRA and SGD accuracy on each of the 128 matched orders. The actual paired SD is 2.40pp; discarding their covariance gives 3.22pp. With Frozen, negative covariance instead increases paired spread.

Two different intervals. The recorded approximate 95% Monte Carlo interval for the controlled MIRA–SGD *mean* is [4.32,5.16]pp; the empirical central order range is [0.45,8.61]pp. The recorded exact interval for its win probability is [93.30,99.51]% (125/128 wins). None is a predictive interval for a new method. The original-order budgets on page 2 are not transported here.

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Distribution checks for the scientist

Check	Reference or expected signature	Failure interpretation
Provider dependence	Preserve joint 4-bit patterns; split all-correct, mixed, all-wrong	Independent provider errors overstate headroom
Paired errors	Report rescue/harm/both-wrong and neutral action changes	A mean gain alone hides offsetting damage
Fitted priors	Use paired five-fit profiles; retain all seed results	Seed duplicates do not add evidence
Delay	No same-decision feedback before selection; matched prefix before release	Early improvement suggests leakage or changed prior
Drift phase	Compare pre/post error trajectory and released affected receipts	A aggregate stress gain may hide slow recovery
Mechanism	Candidate vs its own Frozen and its own permutation	A gain against an old comparator is not attribution
Coverage	Match accepted counts in each fit, not only their sum	More refusal or changed cohort confounds accuracy
Uncertainty	Separate fit variation, new arrivals, new scaffolds, random gate draws	Fixed-trace reweighting is not a new trajectory CI

What has high-confidence support, and what still needs evidence

Verified empirical structure: recorded denominators, provider-error overlap, paired rescue/harm, fit/delay/phase strata, source hashes and table identities. These are finite-cohort facts, not generalization estimates.

Exact conditional requirements: given uniform α/β response, E1 pooled, scaffold and stratum budgets are calculated from the same recorded molecules. E6 utility follows its within-fit swap contract. Conditional validity does not establish attainable selectivity.

Unresolved prediction: no calibrated future outcome distribution, seed SD or predictive CI exists for an unspecified new candidate. Null and adverse effects remain plausible; the previous learned priors actually underperformed. The table should guide which mechanism to repair, not imply certainty of winning.

Before filling a prospective result matrix

Freeze one candidate: features, prior, parameters, update rule, selected-only feedback, release clock, failure semantics and gate. Then match its own Frozen, online baselines and attribution controls. New arrival-order uncertainty requires full adaptive trajectories; unseen-scaffold generalization requires untouched data. The 157 inspected molecules cannot become a new unseen confirmation cohort. Write a calibrated predictive interval only after its model and coverage have been justified.

Evidence and reproducibility

Measured reference: `distribution_audit_v1/` contains 16 CSVs, 19 hashed sources, and an independent standard-library extractor. Actual E1 records include 65,940 method/configuration/molecule rows; E2 includes 28,260. Their Local and Relay terminal vectors match completely. All seed dependencies are retained. `conditional_requirements.json` stores the response assumptions, fractional budget sums, fit/delay profiles, scaffold contrasts, and per-fit swap limits. `build_distribution_guide.py` performs no training, policy replay, model inference or random sampling.

This Guide maps E1–E6 to body Tables 3–8 and Figure 3; the active review standard's E6 untouched-confirmation gate is a different identifier. v01–v05 remain archived; v05 passed arithmetic/visual checks but failed scientific target confidence. Its independently chosen means and smooth delay goals have been superseded. The Guide does not write target values into the real manuscript or measured-result preview.

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